Same Words, Different Bets

Lexicons, static embeddings and contextual pre-training for sentiment classification under scarce labels and ambiguity

How should a small unlabeled corpus be used: static embeddings or contextual pre-training?

SST-2 + IMDb | CPU only | 3 seeds | Anurag Prasad

The problem: eight questions, shared evaluation

Shared splits and tests; model-specific features and tokenizers

RQ1-2	Does pre-training help, and how robust is it?
RQ3-4	How much pre-training is enough, and what does it cost?
RQ5	How far do public checkpoints move the picture?
RQ6	Which method best uses the available unlabeled corpus?
RQ7	How do polarity strength and confidence relate to errors?
RQ8	How well does each model transfer across domains?

Approach: six model families

Shared evaluation sets; model-specific features and training procedures

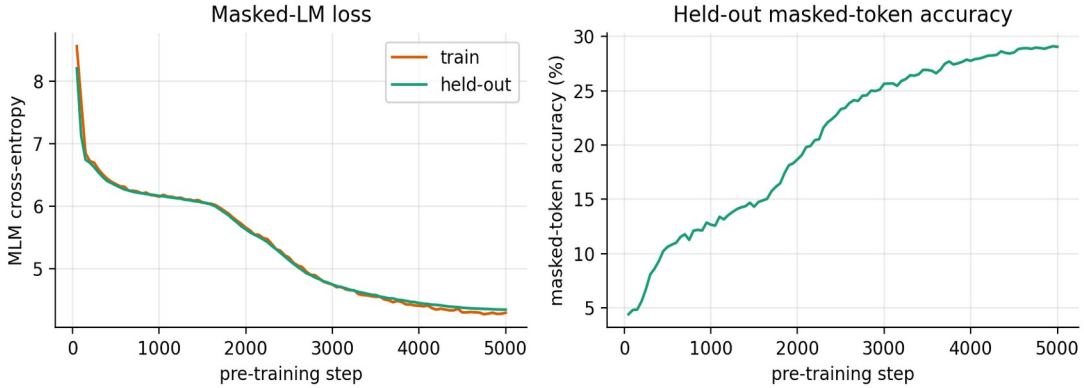
VADER lexicon 0 task labels	TF-IDF word/char + LogReg/NB/SVM	word2vec / fastText same 1.2M words	Tiny-BERT random init	Tiny-BERT +MLM pre-train	BERT-Tiny / DistilBERT public

Evaluated on: clean SST-2, typos (10/30%), word-drop (20/40%), out-of-domain IMDb, and - new - ambiguity strata from fine-grained SST-5 labels (strong vs weak polarity), confidence calibration, and linguistic markers (contrast, negation).

Tiny-BERT: 2 layers, hidden 128, 1,173,248 params. MLM pre-training: 5,000 steps, 13 min, on 1.2M unlabeled IMDb words. Sparse-model regularisation is tuned on dev; the public-checkpoint learning rates are fixed, and embedding-family settings are not exhaustively tuned.

Pre-training starts with a long plateau

Held-out masked-LM loss and accuracy over 5,000 steps



- First ~1,500 steps: loss stuck near the unigram entropy
- Only then does attention start to use context

ppl 77

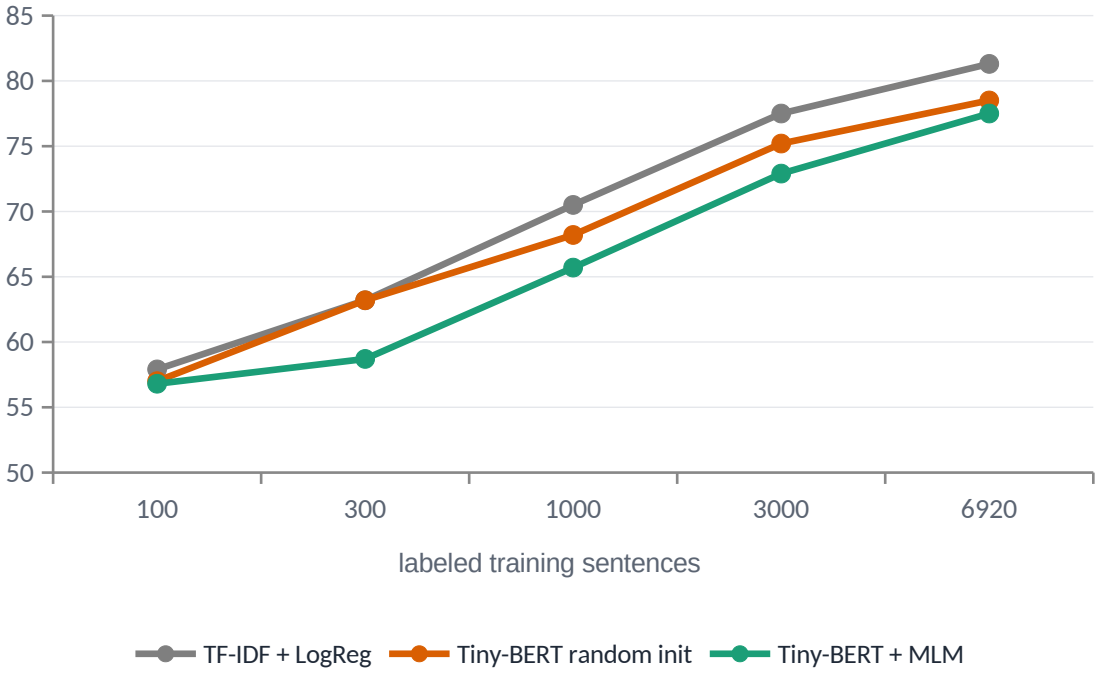
final held-out perplexity (from ~3,660)

29%

masked-token accuracy

Pre-training did not help; TF-IDF wins

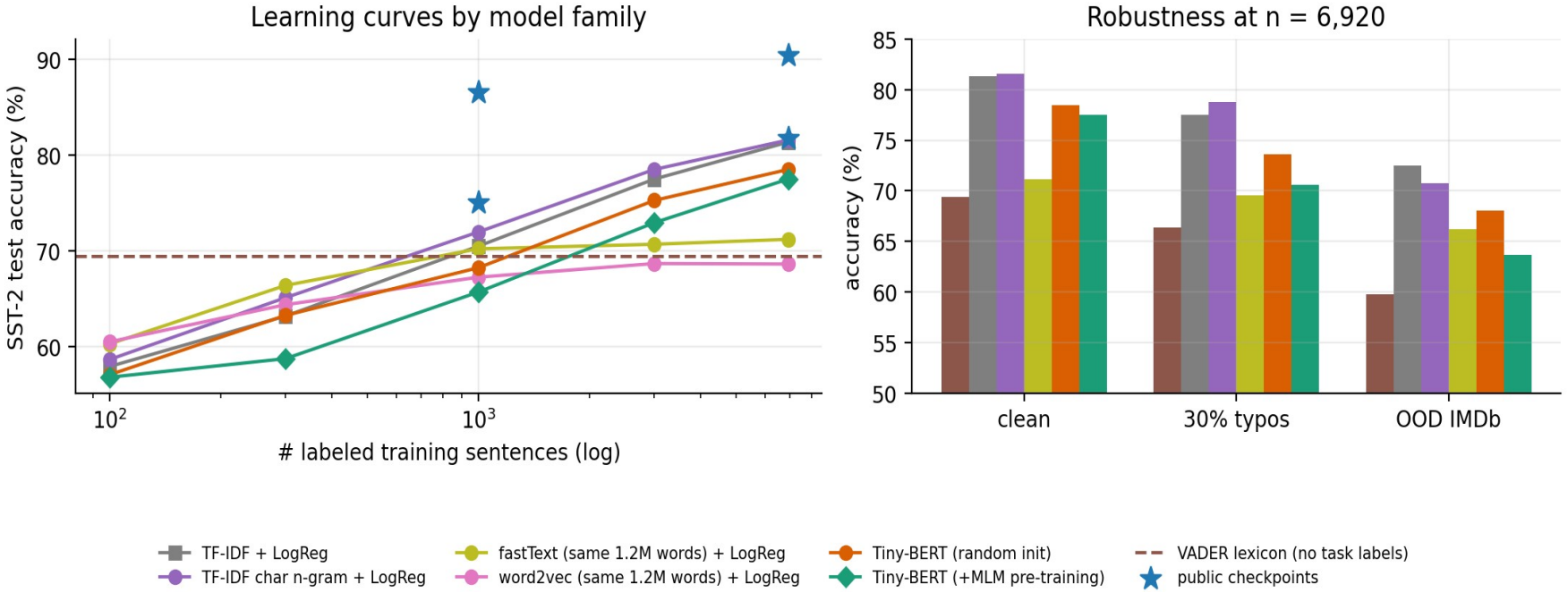
SST-2 test accuracy (%), mean of 3 seeds



- MLM vs random init: -0.3 to -4.5 pts at every budget (CIs exclude 0 at n=300,1k,3k)
- TF-IDF beats the pre-trained model by 3.9-4.8 pts from n=300 on

RQ6: the best bet depends on the label budget

SST-2 accuracy by model family (mean of 3 seeds)



- fastText (same 1.2M words) is the best TRAINABLE model at 100-300 labels; it beats MLM at n=1000 (70.2 vs 65.7), then plateaus at 71.2%
- VADER (0 task labels): 69.4% - worth ~840 TF-IDF labels

A currency for comparison: label-equivalence

How many TF-IDF labels is each model worth? (interpolated on the TF-IDF curve)

Model	n = 1,000	n = 6,920
VADER (0 labels)	~840	--
fastText (same words)	~960	~1,120
Tiny-BERT +MLM (ours)	~450	~3,000
BERT-Tiny (public)	~2,040	>6,920
DistilBERT (public)	>6,920	>6,920

0.45x

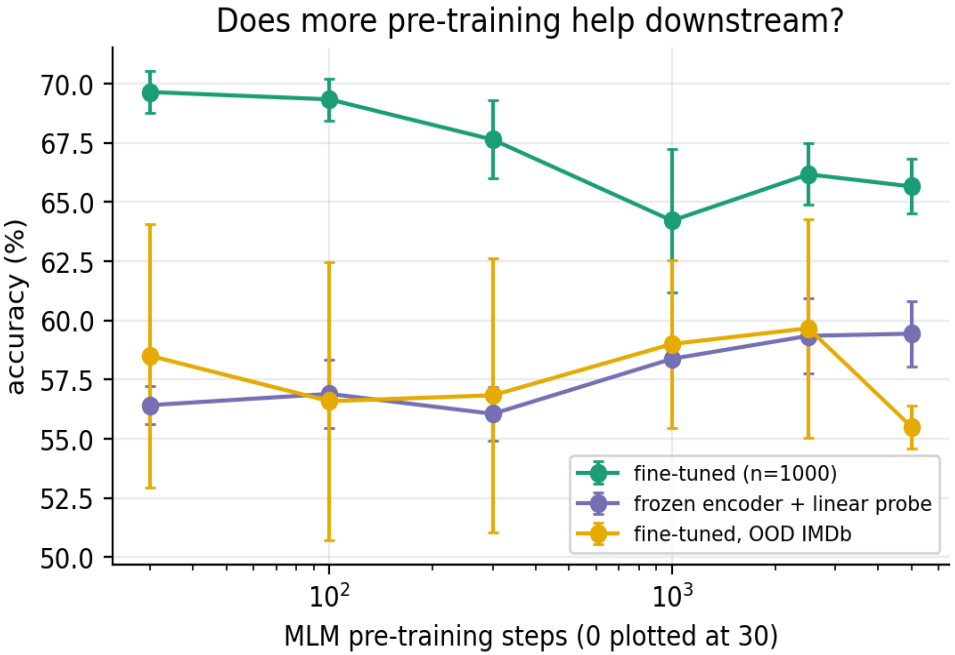
our pre-trained tiny BERT: 1,000 labels ~ 450 TF-IDF labels

6.9x

DistilBERT with 1,000 labels beats TF-IDF with all 6,920

RQ3: the probe improves, fine-tuning does not

Checkpoints after 0-5,000 MLM steps; n = 1,000 labels; 3 seeds



56.4 -> 59.4

frozen linear probe: MLM does store sentiment info

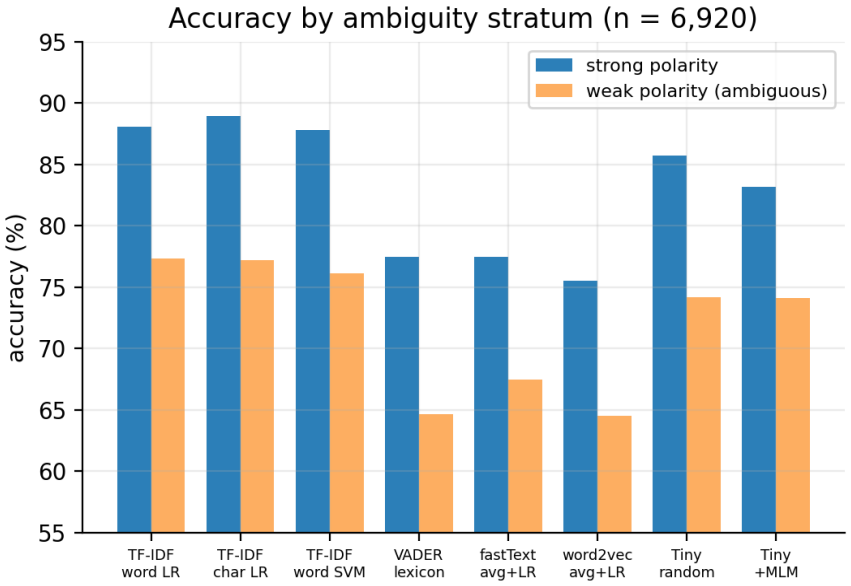
69.7 -> 65.7

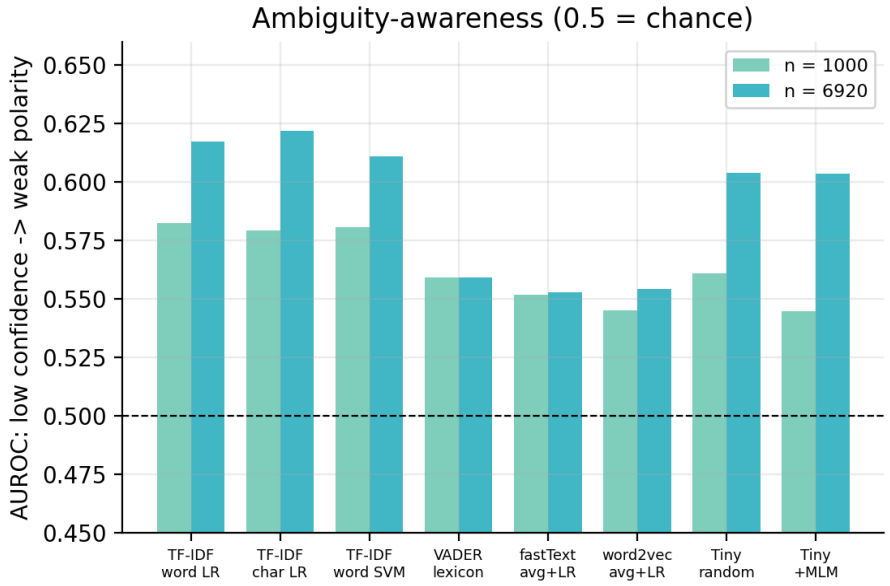
fine-tuned accuracy falls with pre-training

A probe alone would have predicted a benefit.

RQ7: a conditional polarity-strength effect

SST-5 strength is a proxy, not direct annotator disagreement

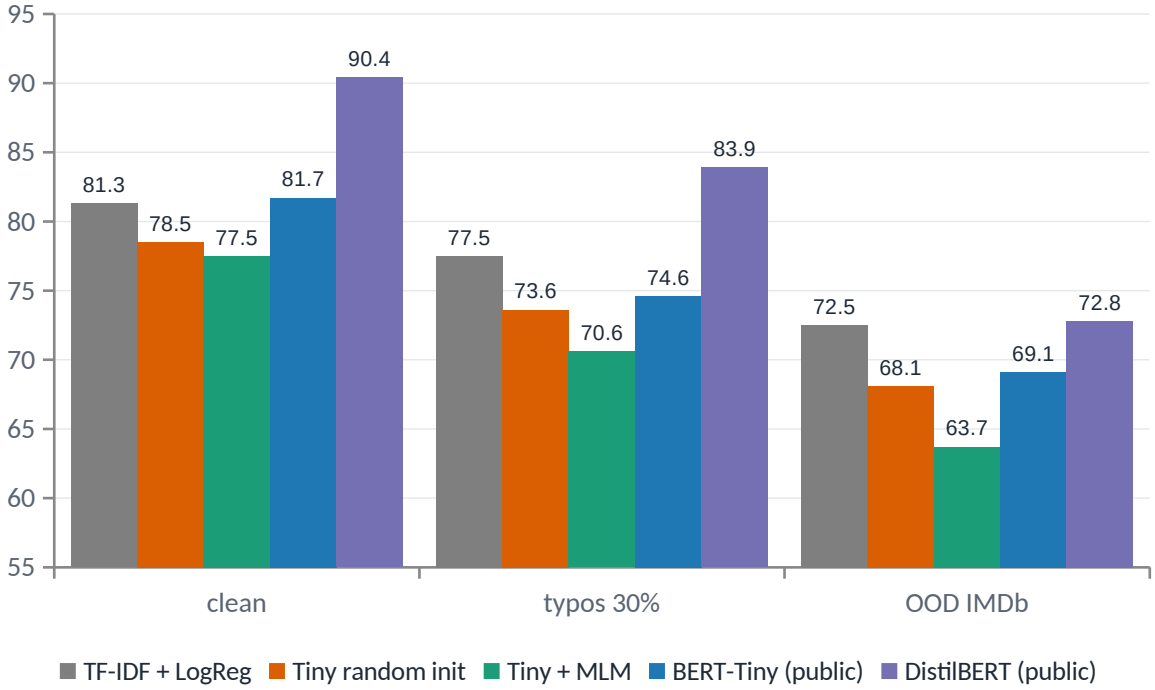




- At n=1000: -4.7 pts on strong-polarity vs -1.3 on weak-polarity examples; weak-stratum CI includes zero
- Best ambiguity-awareness AUROC across ALL models: 0.62 (chance = 0.5) - nobody's confidence tracks ambiguity well
- Transformers on 1,000 labels are badly overconfident: ECE 23-26% vs 3-6% for TF-IDF

RQ2: typos and domain shift

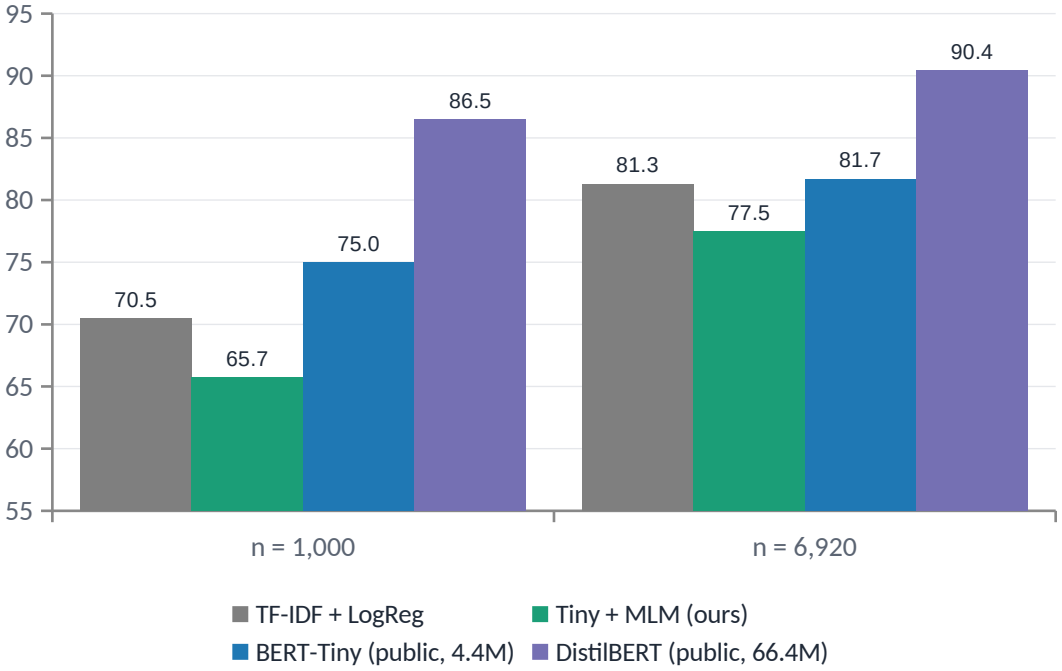
Accuracy (%) at n = 6,920 labeled sentences



- 30% typos: transformers lose 4.9-7.1 pts vs 3.8 for TF-IDF
- OOD: pre-training did not help ours; no public model beat TF-IDF

Public checkpoints perform better

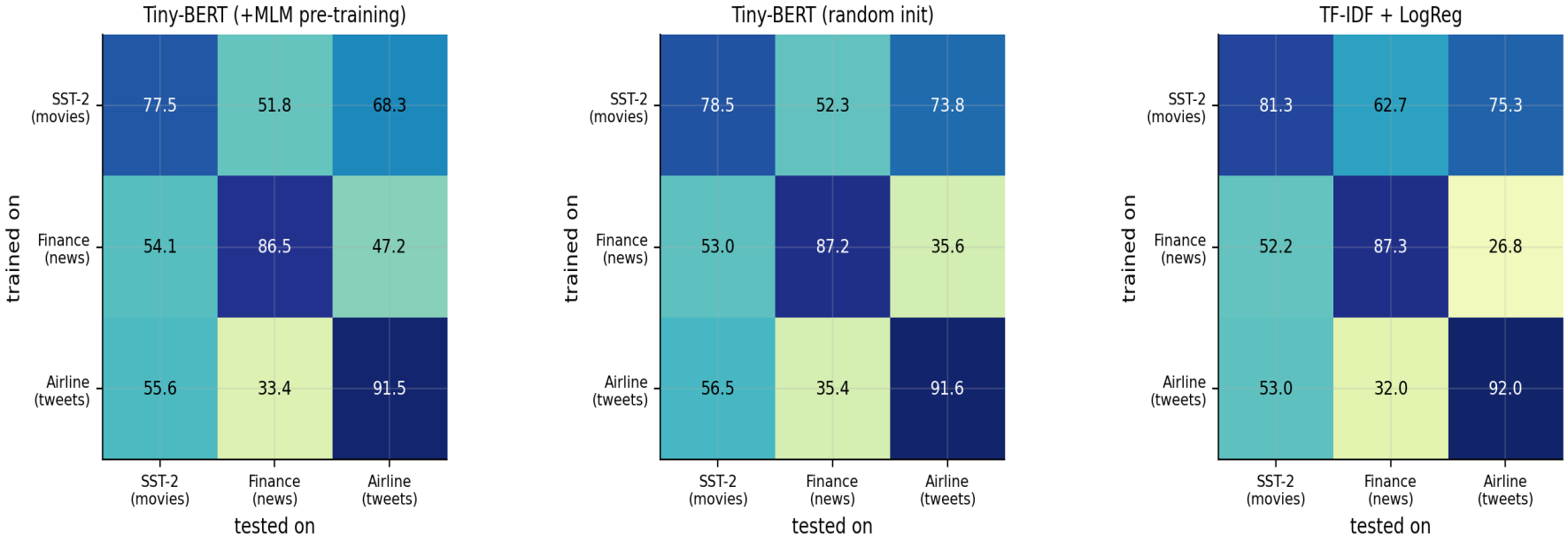
SST-2 accuracy (%); public models differ in corpus, vocabulary and training recipe



- BERT-Tiny has OUR depth and width: +9.3 pts at n=1k over our pre-trained model
- DistilBERT: +16/+9 pts over TF-IDF, ~26 min to fine-tune vs 37 s
- Consistent with a pre-training-scale benefit, but scale is not isolated here

RQ8: does it travel? Cross-domain generalisation

Financial PhraseBank (finance news, real agreement) + Twitter Airline (tweets, real confidence)



- Low-budget results are descriptive: fastText +3.2 pts over MLM at airline n=100; paired intervals need rerun
- Full-data means are close, but no claim of equivalence without reproducible paired tests
- Transfer is directional: scratch Tiny SST-2->airline -4.7 pts; airline->SST-2 -35.1; finance TF-IDF->airline -60.5
- Macro-F1 matters: finance agreement strata drop 6-9 pts; airline confidence strata drop 16-19 pts (signals differ)

Takeaways

Three findings, one field guide

It depends

fastText wins <1k labels; TF-IDF wins after; MLM never does

Where it differs

larger estimated deficit on strong-polarity examples

Be cautious

confidence weakly detects the polarity-strength proxy

- Field guide: few hundred labels -> fastText; few thousand -> TF-IDF; public checkpoint available -> use it
- Public checkpoints perform better, but the comparison does not isolate corpus scale from vocabulary and recipe
- Next: tuned checkpoints, more seeds, larger OOD tests and direct annotator-disagreement labels

BACKUP

Backup: cost of each model

Single CPU thread; seed-0 full-data models

Model	Size (MB)	Latency (ms)	Sent./s	Acc. (%)
Tiny random init, FP32	4.71	1.03	4,914	78.3
Tiny +MLM, FP32	4.71	1.02	5,073	76.1
Tiny +MLM, INT8 dynamic	3.54	1.71	3,196	75.9
TF-IDF + LogReg	2.72	0.31	48,388	81.3
fastText + LogReg (est.)	~0.5	~0.5	~10,000	71.2

- INT8: -25% file size, accuracy unchanged within noise (single model)
- fastText row is an order-of-magnitude estimate, not separately measured

BACKUP

Backup: statistical evidence

SST-2 paired sentence bootstrap; intervals conditional on the saved seeds

Comparison	n=1000	n=6920
+MLM - random init, STRONG sentences	-4.7* [-7.2,-2.2]	-2.6* [-4.8,-0.2]
+MLM - random init, WEAK sentences	-1.3 [-3.4,+0.8]	-0.1 [-2.2,+1.9]
+MLM - random init, TF-IDF LR (overall)	-2.5* [-4.2,-1.0]	-1.0 [-2.6,+0.5]

- Interaction (strong - weak) at n=1000: -3.5 [-6.7,-0.3], CI excludes 0
- Method: resample 1,821 test sentences 2,000 times after averaging paired per-sentence correctness differences over matched seeds; this does not capture retraining variability

Backup: limitations and threats to validity

BACKUP

What the results do and do not show

- Ambiguity proxy is SST-5 polarity STRENGTH, not annotator disagreement; 'weak' covers 62.8% of sentences
- Only 2 public checkpoints (BERT-Tiny, DistilBERT), untuned learning rates, predictions not stored (no paired tests for them)
- Embedding families untuned (fixed dim/epochs/pooling); no CNN/BiLSTM baseline
- 3 seeds; full 872-sentence dev set for early stopping even at n=100
- One task and one language (English sentiment), across three domains

Backup: reproducibility

Three-domain base study; public checkpoints optional

```
bash run_all.sh              # clean run of SST-2 + finance + airline
python run_experiments.py --stage families # VADER,SVM,char-gram,word2vec/fastText
python ambiguity.py          # SST-5 strata,AUROC,ECE
python hf_benchmark.py --models ladder --tune # size ladder,tuned lr
```

- Resumable runs and saved predictions; paired SST-2 tests plus an explicit optional multi-domain bootstrap
- Deliverables: code, technical docs, main report, 2-page report, ACM paper, this deck